

Statistical Analysis of Temperature and Rainfall Trends in Adelaide (1995 – 2026)

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1. Abstract

The study aims at identifying the relationship that exists between the trends in temperature and rainfall within Adelaide of South Australia using the climatic data gathered over the years 1995 to 2026 from Bureau of Meteorology (BOM) Adelaide Airport Station. In particular, this research focuses on identifying any possible climatic trends in order to determine whether increase in temperature leads to decrease in rainfall. This was achieved through various statistical techniques such as Univariate Analysis, Correlation analysis, Scatter diagram, and Linear Regression. Consequently, there was found to exist convincing evidence of negative statistically significant correlation between temperature and rainfall. Based on the Spearman Rank correlation, there existed a moderate negative relationship between the two variables and the linear regression showed that there exists a decrease in rainfall of about 3.24mm when temperature increases by 1° C. As such, Adelaide is experiencing a warming and drying trend that can be an issue of concern to water resource management, climate adaptation measures as well as environmental planning. Additionally, this study compares its findings with those conducted previously in South Australia and highlights the importance of statistical analysis to recognize regional climate variability.

2. Keywords

Climate trends
Rainfall Analysis
Temperature Trends
Statistical Analysis
Spearman correlation
Regression Analysis

3. GitHub Repository

https://github.com/Unicorn-EDA/Temperature_Rainfall_Analysis_Adelaide/tree/main

4. Introduction

Climate change has become a central theme in discussions on environment and engineering in the twenty-first century. It is the warming of the Earth, the changes in rainfall and other climate features that have been identified as the major causes of urban sustainability issues, water scarcity, farming systems, and infrastructure design all over the planet today. Australia has been receiving climate reports on many occasions which indicate that there are several areas that either temperature is rising or rainfall lessening. The South Australian part of the country is among the places experiencing these environmental changes. Temperature and rainfall are linked very tightly in the hydrological cycle. If temperature fluctuates, it can influence the levels of evaporation, the storage capacity of moisture in the atmosphere, and the amount of rainfall. Knowing the mathematical formula connecting the two factors becomes essential for wanting to forecast environmental conditions and making planning decisions in the future.

Various studies have investigated the relationship between rainfall and temperature variations or vice versa. According to Ye and Ahammed (2020), the results in their study derived from the examination of the relationship between the maximum daily temperature and maximum daily rainfall in South Australia using non-parametric statistics such as Kruskal Wallis test, Mann-Whitney test, and Spearman's Rank Correlation test. The two authors showed that both rainfall and temperature were weather station-dependent and there was no significant correlation between the two. In the same way, Murphy and Timbal (2008) reported the occurrence of low rainfall and high temperatures in southeastern Australia.

Despite earlier investigations into climatic variability for larger areas within Australia, there is no extensive literature on temperature and rainfall relationship in Adelaide for a long duration by trend analysis and modelling. This investigation bridges the gap by using climate data for Adelaide, sourced from the Adelaide Airport Bureau of Meteorology Station, for a period of 30 years between 1995 and 2026.

Major objectives of this study include:

- Analysing long-term temperature and rainfall patterns of the city of Adelaide.
- Statistical analysis of the association between temperature and rainfall using bivariate analysis.
- Determining whether high temperature is associated with low rainfall.

5. Data

The data used was taken from the BOM data set where the station selected is Adelaide Airport weather station (Station number 23034). Our choice of Adelaide Airport Station was because there was availability of data spanning many decades and thus representing the climatic conditions of the Adelaide Metropolitan Area. The data spanned more than 30 years from March 1995 to February 2026.

Below are the two main climate variables analysed:

- Average Daily Maximum Temperature (°C)
- Rainfall (monthly) (mm)

Temperature data show the average maximum temperatures registered each day at the station. Rainfall data refer to the monthly totals of precipitation in millimetres (mm). Firstly, it is important to start working with the basic dataset and its preprocessing. This involves removing potential errors, inaccuracies, and inconsistencies from the dataset, fixing all missing data, duplicates, etc. Secondly, after preprocessing, sorting of data according

to time periods is carried out and the data are loaded into the statistical analysis software for visualization purposes and further analysis.

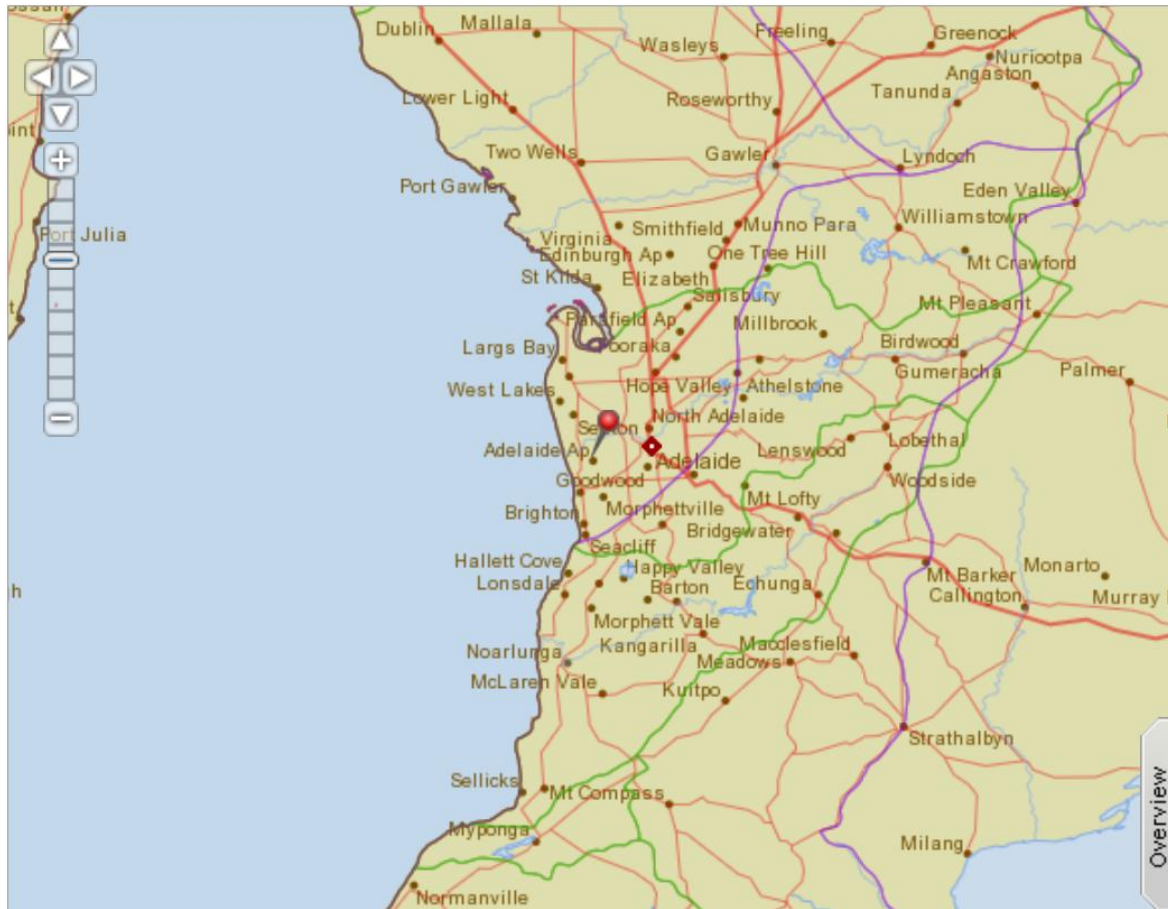


Figure 1 Map of Study Area

Table 1 Dataset Summary

Variable	Unit	Time Period	Source
Mean Maximum Temperature	°C	1995 – 2026	BOM
Monthly Rainfall	mm	1995 - 2026	BOM

6. Methods

The analysis of various statistical methods like univariate analysis, bivariate analysis, correlation analysis, and linear regression was the main tool of this present research study to identify the pattern of temperature and rainfall in Adelaide.

6.1 Univariate Analysis

First, univariate statistical analysis was performed independently for the temperature and rainfall variables to understand the character and distribution of the individual data points. The character and distribution of the data were assessed by visualization tools like histograms and QQ plots. Histogram analysis was a key method in deciding the type of distribution of frequency and distribution of values in a data set. Meanwhile, QQ plots were the method in assessing the distribution of the data set by checking it against that of a theoretical normal distribution.

6.2 Bivariate Analysis

Bivariate analysis done to establish the relationship between rainfall and temperatures. The scatter plot was used to discover the nature, direction, and level of relationship. Looking at the scatter plot the relationship is negatively correlated because high temperatures are related to low rainfall.

6.3 Correlation Analysis

The Spearman Rank Correlation was used to test statistically whether there is any relationship between temperature and rainfall. The reason for using Spearman correlation was that the two climatic variables under analysis did not follow a normal distribution. A significance level of 0.05 was used to determine statistical significance.

The hypotheses used in this study were:

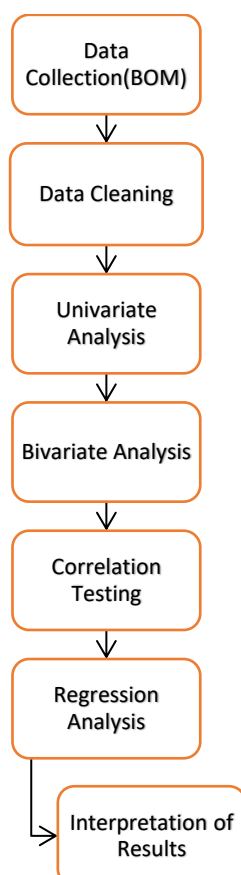
Null Hypothesis (H_0): There is no statistically significant relationship between temperature and rainfall in Adelaide.

Alternative Hypothesis (H_1): There is a statistically significant relationship between temperature and rainfall in Adelaide.

6.4 Regression Analysis

In an attempt to determine the effects of temperature on rainfall and to determine the relationship between these two variables, simple linear regression technique was applied.

A flowchart depicting the method is shown below:



7. Result

7.1 Univariate Analysis Results

A statistical distribution of temperature and rainfall has been checked through histograms and Q-Q plots. Temperature statistics are not normally distributed. It has been theoretically proved using a Q-Q plot that the temperatures follow the pattern of a not normal distribution.

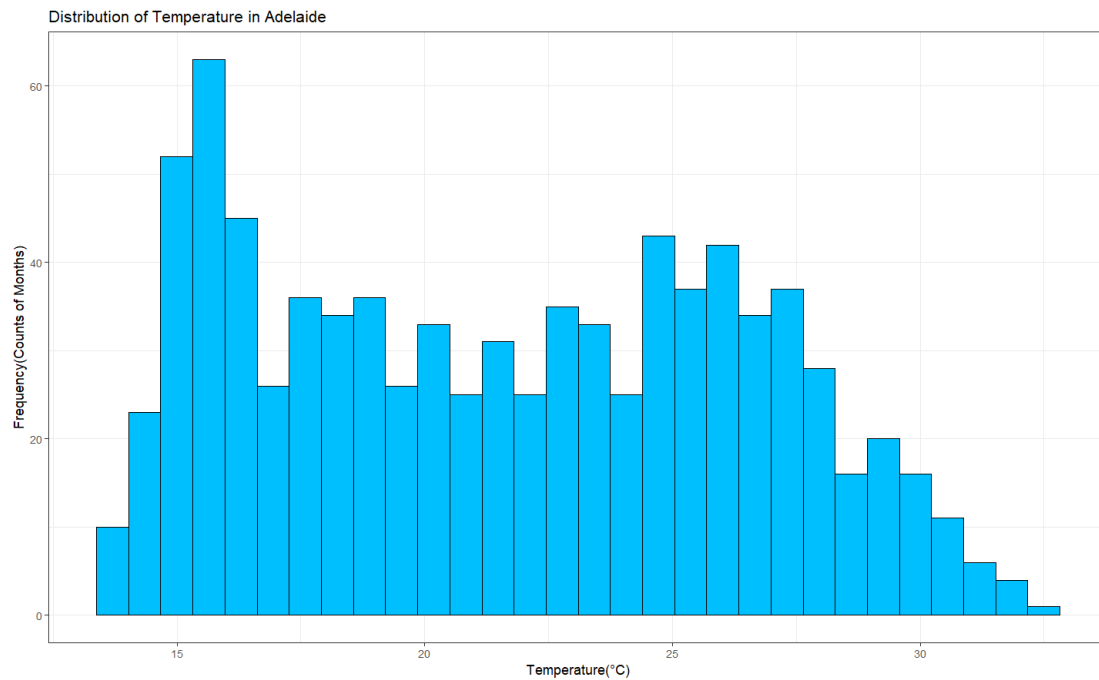


Figure 2 Histogram of Temperature

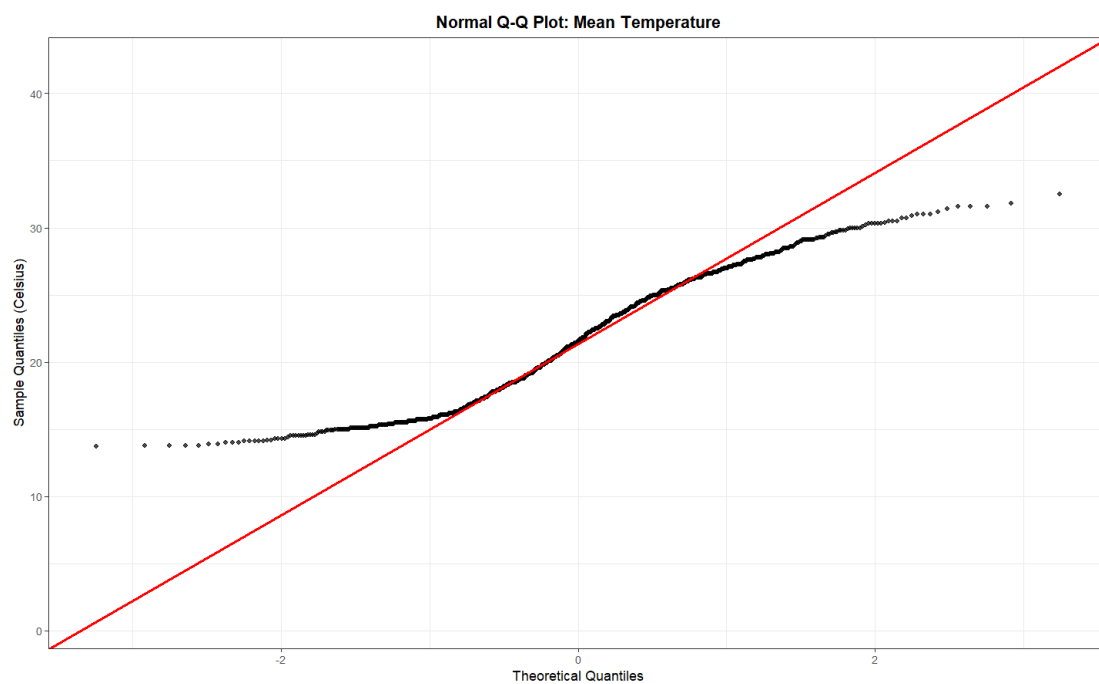


Figure 3 Q - Q Plot of Temperature

The rainfall had a rather irregular or skewed distribution pattern. There were instances of very heavy as well as very light rainfall, which resulted in a very high variation in the rainfall data set. This skewness or asymmetry was also noted from the rainfall Q-Q plot.

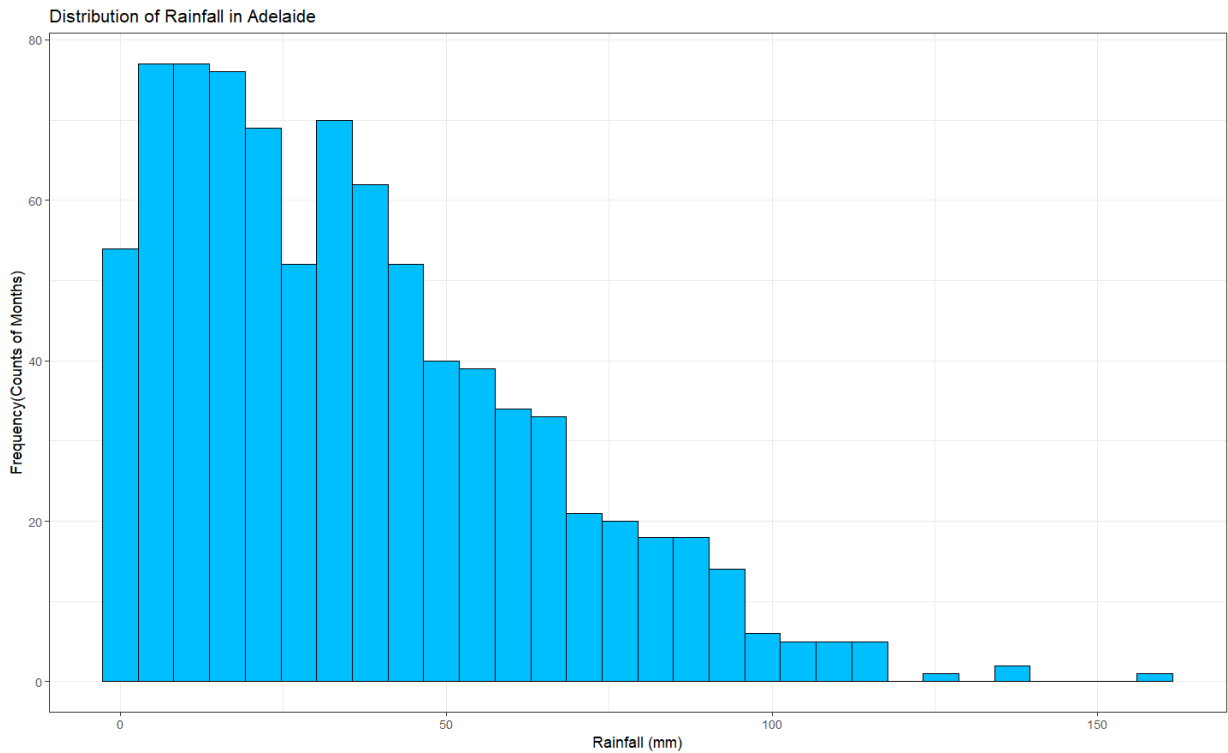


Figure 4 Histogram of Rainfall

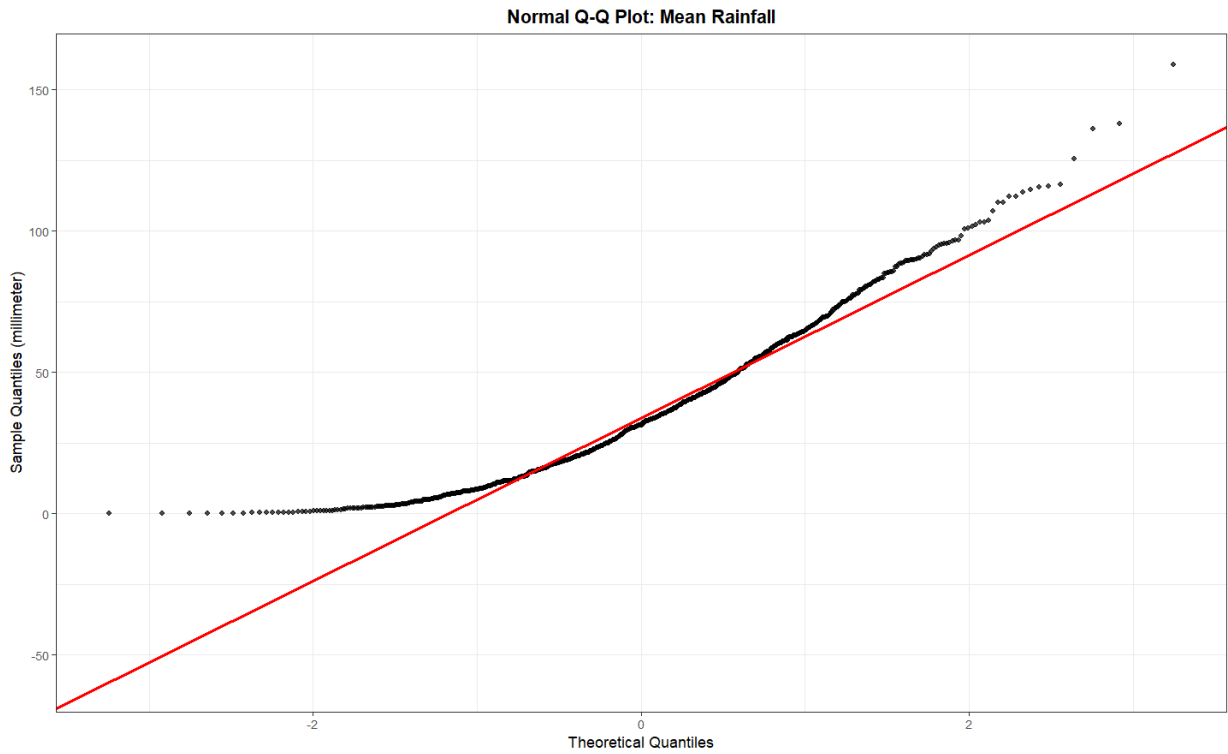


Figure 5 Q - Q Plot of Rainfall

7.2 Bivariate Analysis Results

Based on scatter diagrams of bivariate analysis, it can be observed that there is negative correlation between the variables, which means that with increase in the level of one variable, there is a decrease in the value of another one. The current trend shows that higher temperature is associated with lower rainfall in Adelaide. The scatter plot showed that the data distribution pattern was also relatively scattered compared to the trend line, implying that the data can also be influenced by other climate factors.

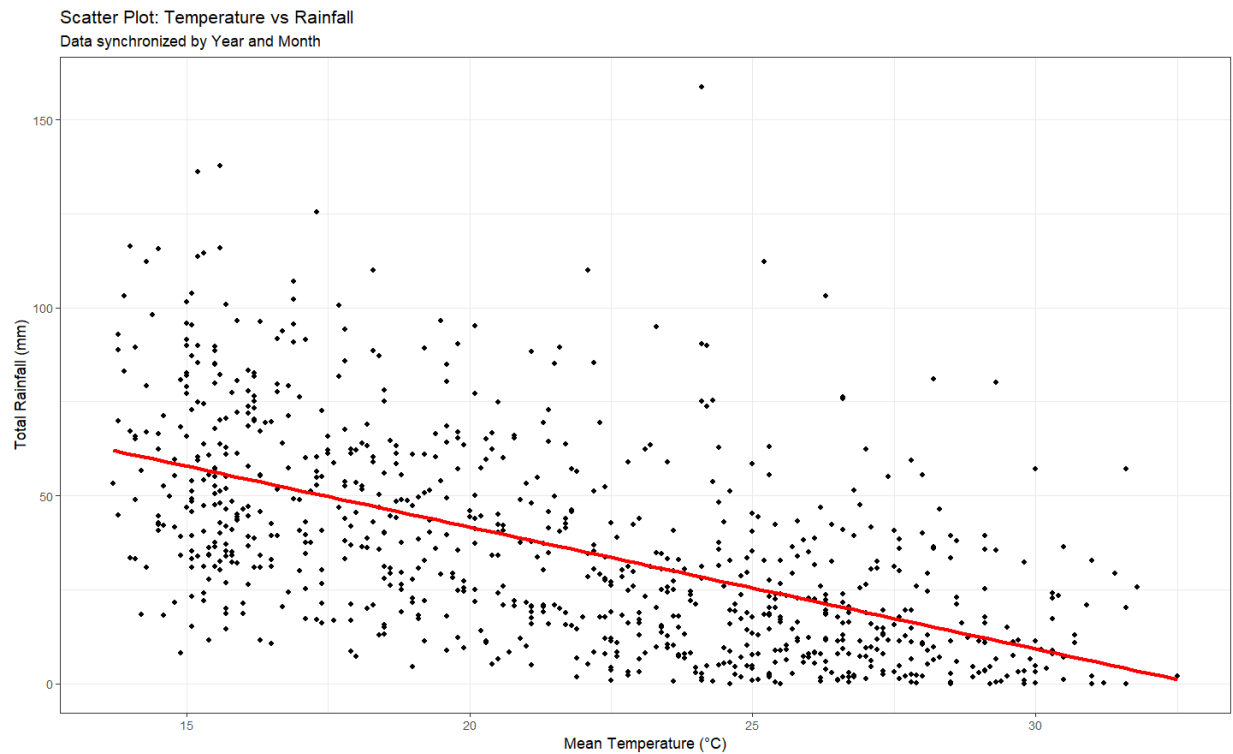


Figure 6 Scatter Plot

7.3 Correlation Analysis Results

The Spearman's Rank Correlation was used to analyse the correlation between temperature and rainfall. The results showed that the relationship between temperature and rainfall is negative. The p-value produced was less than 2.2×10^{-16} , which is way less than the significance value of 0.05. Thus, it should be noted that reject the null hypothesis and accept the alternative hypothesis.

Table 2 Correlation Analysis Results

Test	Correlation Coefficient	P-Value	Interpretation
Spearman's Rank Correlation	-0.607	<2.2e-16	Significant negative relationship

The findings of research suggest that there is a moderate correlation between high temperatures and low rainfall in Adelaide. Contrary to what is stated in the research mentioned by Ye and Ahammed (2020), which concludes that South Australia cities have poor relationships with each other, it can be said that there is a negative correlation in Adelaide.

7.4 Regression Analysis Results

In order to examine how temperature affects rainfall, linear regression analysis was conducted. The equation for regression is as follows:

$$\text{Rainfall} = 106.51 - 3.24(\text{Temperature})$$

Table 3 Regression Analysis Results

Parameter	Value
Intercept	106.51
Slope	-3.24
R ²	0.32
P - Value	<0.05

The slope coefficient coming from the regression equation reveals that if the temperature increases by 1 degree, the rainfall will decrease by nearly 3.24mm. Based on the r^2 value of 0.32 that was obtained, one can deduce that temperature is responsible for around 32% of the variation in rainfall. It may not be fully explaining the changes in rainfall, but it definitely shows that there is a relation between rainfall and temperature. The other differences could be due to other environmental and atmospheric factors that were not considered in this study.

7.5 Regression Diagnostic Analysis

Residual diagnostic plots were examined to determine whether the regression model assumptions were valid. The distribution of residuals was quite random around zero, which means that the linearity and the behaviour of variance could be considered as acceptable. The Q-Q plot indicated that residuals were roughly normally distributed, Despite some slight deviations at the tails. The leverage analysis showed that there were no strong influential outliers.

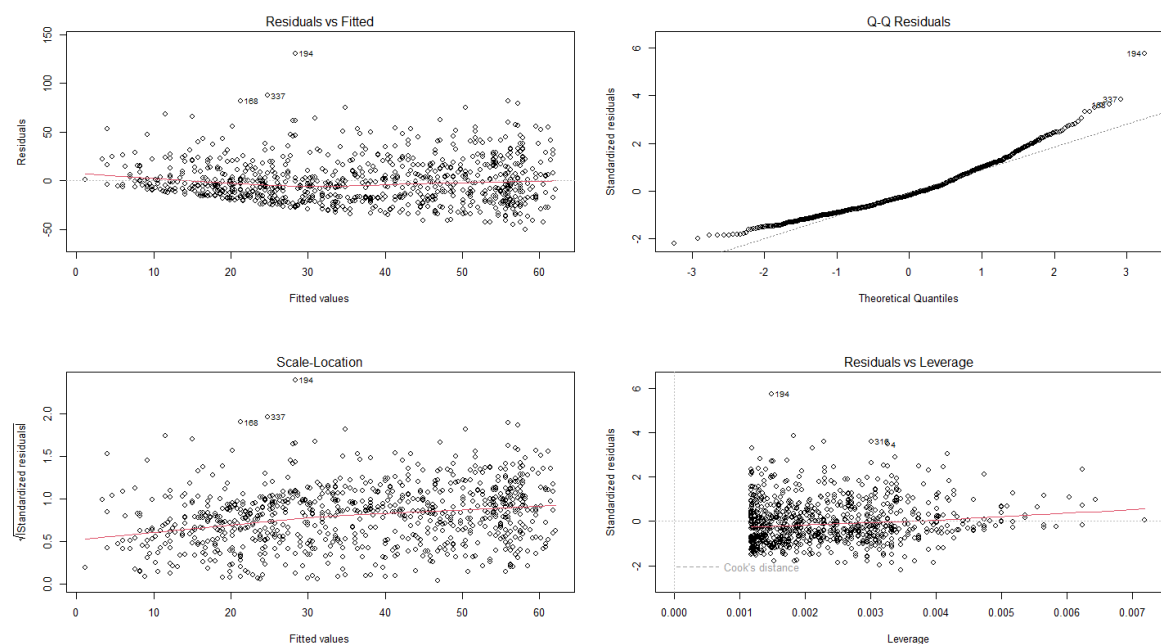


Figure 7 Residuals Diagnostic Graphs

8. Conclusion

The research exploring how rainfall and temperature in Adelaide affect each other with a time series examination of changing climate characteristics. From the research, it was deduced that only two climate variables temperature and rainfall are yet inversely correlated which means generally when temperature rises, rainfall decreases. Because of this, the findings are in line with the concept that a climate change trend involving the warming and drying of Adelaide is actually occurring. Besides, the result of the linear regression analysis reveals that rainfall will reduce by 3.24 mm with each increase of 1 degree centigrade in temperature and temperature accounts for nearly 32% of rainfall variations.

Moreover, after analysing these variables, it came to our notice that rainfall was significantly dynamic as compared to the temperature due to highly complex climatic conditions in South Australia. The correlation analysis indicated that there was moderate negative correlation between the two variables. This report may provide valuable insights to the environmental planners, engineers who want to study about the effect of climate variations on the water resources of the region. For more precise climatic predictions and rainfall forecast, other environmental parameters such as humidity, evaporation rate, wind speed and atmospheric pressure should be considered. The use of ARIMA models and multivariate regression for prediction will prove helpful.

9. Reference

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